

Lecture 2

Game Design



References

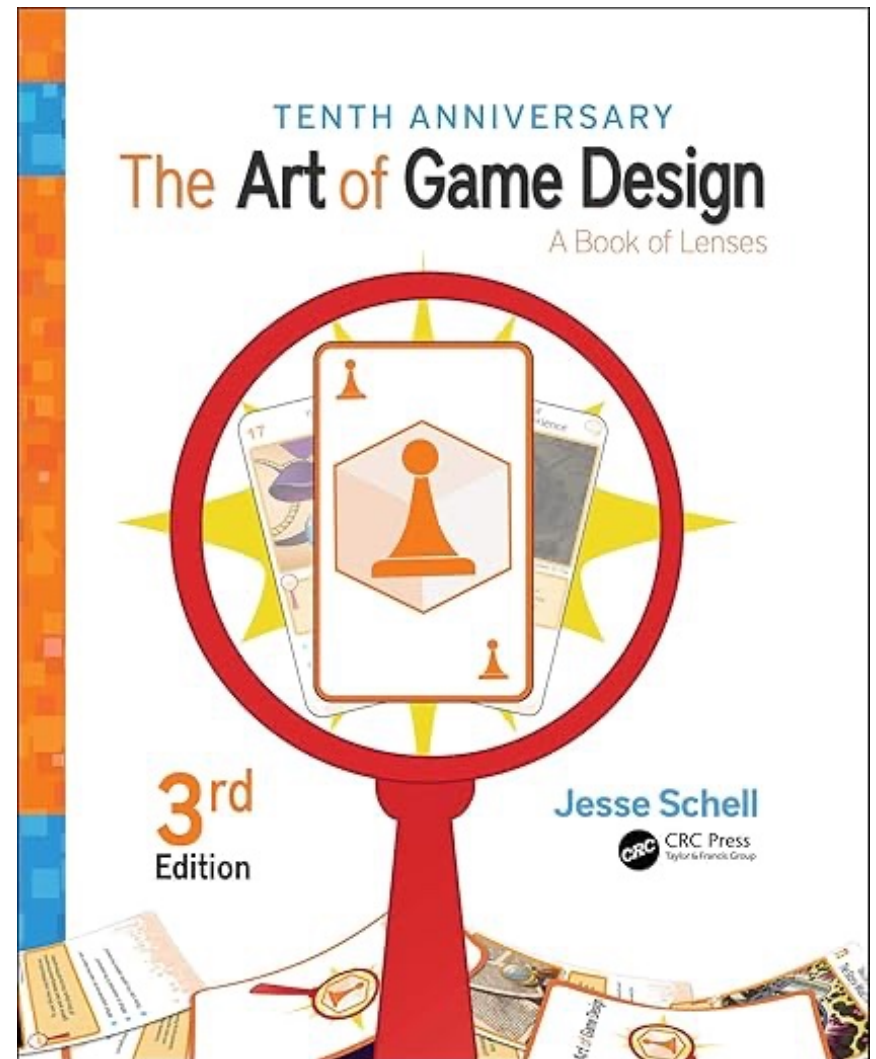
The Art of Game Design

A Book of Lenses

3rd edition

Jesse Schell, 2019

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What is Game Design?

- *“Game design is the act of deciding what a game should be. That’s it. On the surface it sounds too simple.”*
- *“Good game design happens when you view your game from as many perspectives as possible. I refer to these perspectives as lenses, because each one is a way of viewing your design. The **lenses are small sets of questions** you should ask yourself about your design.”*

in Jesse Schell, The Art of Game Design: A Book of Lenses (2008)

<https://deck.artofgamedesign.com/#/?lang=en>

Designing an Experience

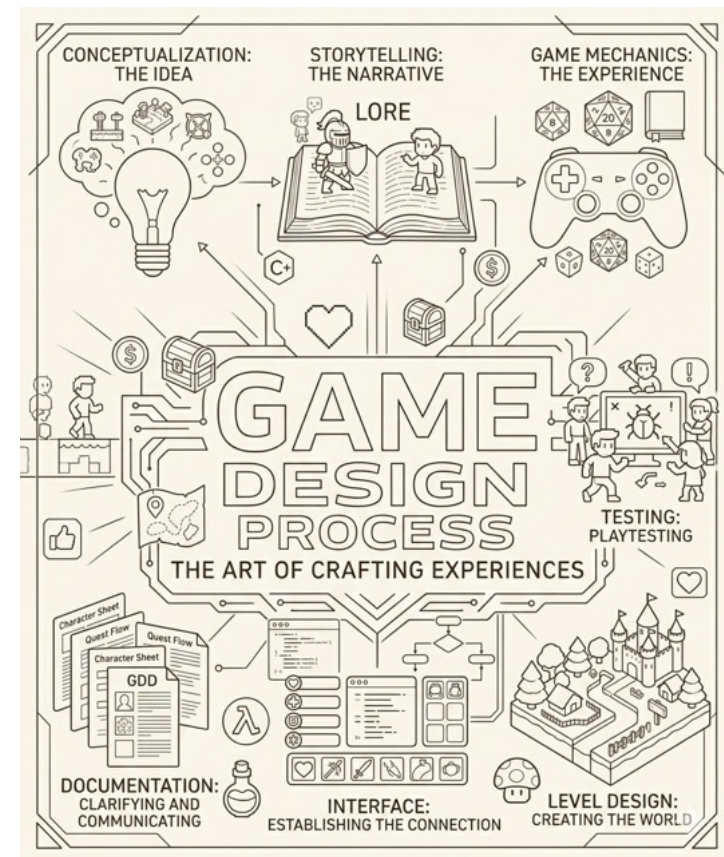
- The game designer creates the experience
 - How to create good experiences?
 - This encompasses: **Psychology**, **Anthropology** and **Design**.
 - *“The only reality that we know is the reality that we experience. We filter reality through our senses and through our minds, and the consciousness we actually experience is a kind of illusion.”*
- Game designers must have a lot of skills, but the most important is to listen:
 - Team, audience, game, client and self.



[Theme Hospital, 1997]

Game Design

- **Conceptualization: The Idea**
- Storytelling: The narrative
- Game mechanics: the Experience
- Level Design: Creating the World
- Interface: Establishing the Connection
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Conceptualization

Emotion

- What emotions would I like my player to experience? Why?
- What emotions are players (including me) having when they play now? Why?
- How can I bridge the gap between the emotions players are having and the emotions I'd like them to have?



Essential Game Experience

- What experience do I want the player to have?
- What is essential to that experience?
- How can my game capture that essence?



Game Surprise



- What will surprise players when they play my game?
- Does the story in my game have surprises? The game rules? Artwork? Technology?
- Do your rules give players ways to surprise each other?
- Do your rules give players ways to surprise themselves?

A hand holding a magnifying glass over a box containing text.

*Lens #4:
The Lens of
Surprise*

Fun and Curiosity

- What parts of my game are fun? Why?
- What parts need to be more fun?



- What questions does my game put into the player's mind?
- What am I doing to make them care about these questions?
- What can I do to make them invent even more questions?



Problem Solving

*Lens #8:
The Lens of
Problem Solving*

- What problems does my game ask the player to solve?
- Are there hidden problems to solve that arise as part of gameplay?
- How can my game generate new problems so that players keep coming back?



[Portal, 2007]

Where to start?

- Get an idea, choose a genre and a visual style
- Brainstorm the idea with the team without restrictions
- Find a story: focus the attention of the team
- Develop characters:
 - Makes story more real and credible
 - Humanize characters
 - Define roles for the characters
- Accommodate restrictions only in the end (publisher/marketing, genre...)



Where to start?

- The game development process is iterative.
- Prototyping is very important to mitigate risks.
- Sometimes is better to create the toy first and the game later.

Genre and Style

- Genre:
 - Platform, Shooter, RPG, Tower Defense, Strategy, ...
- Style:
 - Top-down, Isometric, Slide-scrolling, Mode7/Faux 3D
 - Real-Time, Turn-based, Time-trial
 - Single-player, Dual-player, multiplayer
 - Online vs. local.

The eight filters

- Does the game feel right?
- Will the intended audience like this game enough?
- Is this a well-designed game?
- Is this game novel enough?
- Will this game sell?
- Is it technically possible to build this game?
- Does this game meet our social and community goals?
- Do the play testers enjoy this game?



Build the Toy First

- If my game had no goal, would it be fun at all? If not, how can I change that?
- When people see my game, do they want to start interacting with it, even before they know what to do? If not, how can I change that?



The player

- In general, what do they like?
- What don't they like? Why?
- What do they expect to see in a particular game?
- If I were a player what would I expect to see?
- What would they like/dislike about my game in particular?

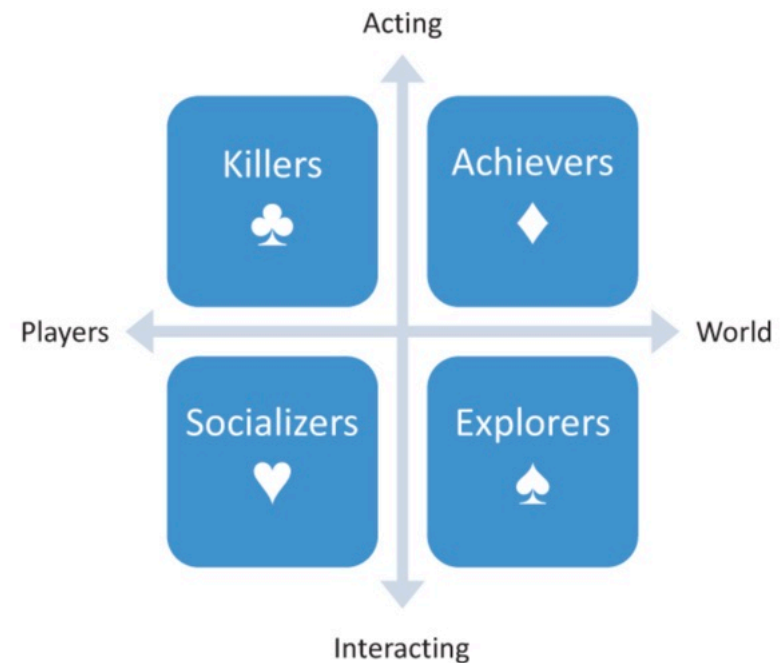


LeBlanc's Taxonomy of Game Pleasures

- **Sensation:** Using your senses to sense the game world.
- **Fantasy:** Imaginary world.
- **Narrative:** The dramatic unfolding of a sequence of events.
- **Challenge:** one of the core pleasures, related to problem solving.
- **Fellowship:** Cooperation and communities.
- **Discovery:** Exploring the game world and discovery of a secret feature.
- **Expression:** The player expresses himself by creating things.
- **Submission:** The suspension of disbelief.

Richard Bartle's Taxonomy of Player Types

- **Achievers** want to achieve the goals of the game: Their primary pleasure is **challenge**.
- **Explorers** want to get to know the breadth of the game: Their primary pleasure is **discovery**.
- **Socializers** are interested in relationships with other people: They seek pleasures of **fellowship**.
- **Killers** are interested in competing with and defeating others: They enjoy a mix of pleasures of **competition** and **destruction**.



From: The Art of Game Design, 3rd Edition
Jesse Schell

Motivation

*Lens #23:
The Lens of
Motivation*

- What motivations do players have to play my game?
- Which motivations are most internal? Which are most external?
- Which motivations are pleasure-seeking? Which are pain-avoiding?
- Which motivations support each other?
- Which motivations are in conflict?



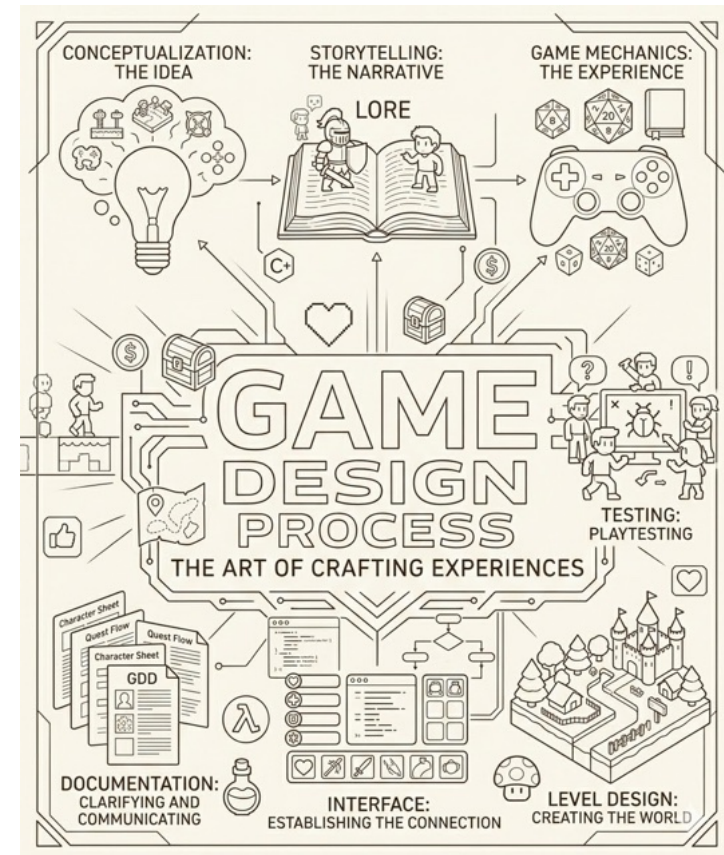
[FarmVille, 2009]

Conceptualization

- **Who** is the game for?
- **What** is the idea, main concept, main characters?
- **What** will be the technology used?
- **What** kind of genre and aesthetics will it follow?
- **Why** will the players be motivated to play?
- **How** will players play the game?

Game Design

- Conceptualization: The Idea
- **Storytelling: The narrative**
- Game mechanics: the Experience
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Storytelling

Stories

- Donkey Kong (1981) was the first to use a visual story line.
- In digital games stories can be non-linear.
- This can generate a combinatorial explosion.
- Multiple endings can disappoint.
- Not enough verbs
Videogame verbs: run, shoot, jump, climb, throw, cast, punch, fly, ...
Movie verbs: talk, ask, negotiate, convince, argue, shout, plead, complain...
- Time travel makes tragedy obsolete.



[The Elder Scrolls III: Morrowind, 2002]

Stories

- Stories can be told in several ways:
 - **Text:** Colossal Cave
 - **Graphic Adventure:** Monkey Island
 - **Cut scenes:** Starcraft
 - **Quick-Time events:** short control between cut scenes: Dragon's Lair (1983)
 - **Scripted Game Scenes:** player loses control temporarily and goes to an animated sequence: Half Life, Resident Evil.



Story Machine



*Lens #73:
The Lens of the
Story Machine*

- When players have different choices about how to achieve goals, new and different stories can arise. How can I add more of these choices?
- Different conflicts lead to different stories. How can I allow more types of conflict to arise from my game?
- Good stories have good interest curves. Do my rules lead to stories with good interest curves?
- A story is only good if you can tell it. Who can your players tell the story to that will actually care?

Different Stories and Plots

- Goat Simulator: <https://www.youtube.com/watch?v=wG7G7kzyAtl>
- Theme Hospital: https://www.youtube.com/watch?v=Le_znuXcP2M
- The Legend of Zelda: <https://www.youtube.com/watch?v=ofGeJV6xIHw>
- Best Stories: https://www.youtube.com/watch?v=1RE_OHe0E74
- Weird Games: <https://www.youtube.com/watch?v=bfB-e8E-8qQ>
- Political Games: https://www.youtube.com/watch?v=S_N3ts77ZDo

Story Tips (I)

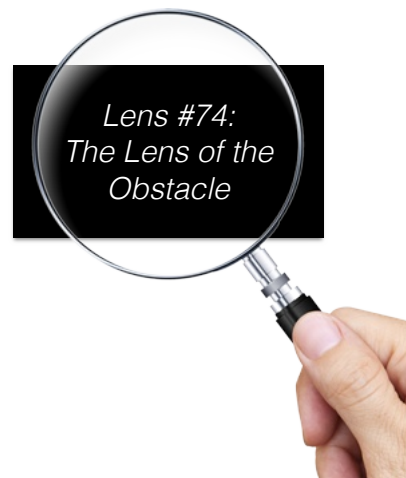
1. Define **Goals**, **Obstacles** and **Conflicts**.
2. Provide **Simplicity**: the game world is simpler than the real world;
and **Transcendence**: the player is more powerful than the real world.
3. Consider the **Hero's Journey**...



[The Legend of Zelda Series, 1986...]

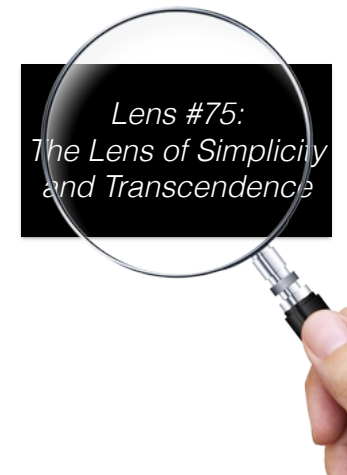
Obstacles

- What is the relationship between the main character and the goal? Why does the character care about it?
- What are the obstacles between the character and the goal?
- Is there an antagonist who is behind the obstacles? What is the relationship between the protagonist and the antagonist?
- Do the obstacles gradually increase in difficulty?
- Some say “the bigger the obstacle, the better the story.” Are your obstacles big enough? Can they be bigger?
- Great stories often involve the protagonist transforming in order to overcome the obstacle. How does your protagonist transform?



Simplicity and Transcendence

- How is my world simpler than the real world? Can it be simpler in other ways?
- What kind of transcendent power do I give to the player? How can I give even more without removing challenge from the game?
- Is my combination of simplicity and transcendence contrived, or does it provide my players with a special kind of wish fulfillment?



Vogler's Synopsis of the Hero's Journey (I)

- **The ordinary world**—Establishing scenes that show our hero is a regular person leading an ordinary life.
- **The call to adventure**—The hero is presented with a challenge that disrupts their ordinary life.
- **Refusal of the call**—The hero makes excuses about why they can't go on the adventure.
- **Meeting with the mentor**—Some wise figure gives advice, training, or aid.
- **Crossing the threshold**—The hero leaves the ordinary world (often under pressure) and enters the adventure world.
- **Tests, allies, enemies**—The hero faces minor challenges, makes allies, confronts enemies, and learns the workings of the adventure world.
- **Approaching the cave**—The hero encounters setbacks and needs to try something new.
- **The ordeal**—The hero faces a peak life or death crisis.
- **The reward**—The hero survives, overcomes their fear, and gets the reward.
- **The road back**—The hero returns to the ordinary world, but the problems still aren't all solved.
- **Resurrection**—The hero faces a still greater crisis and has to use everything they have learned.
- **Returning with the elixir**—The journey is now well and truly complete, and the hero's success has improved the lives of everyone in the ordinary world.

Hero's Journey

- Does my story have elements that qualify it as a heroic story?
- If so, how does it match up with the structure of the hero's journey?
- Would my story be improved by including more archetypical elements?
- Does my story match this form so closely that it feels hackneyed?



Story Tips (II)

...

4. Put your Story to Work! Test in prototypes with any of the elements of the game tetrad.
5. Keep your story world consistent.
6. Make your story world accessible: if necessary, simplify the story to reach a larger audience.
7. Use clichés judiciously: balance familiarity with repetitiveness.
8. Sometimes a map brings a story to life!

The Story

- Does my game really need a story? Why?
- Why will players be interested in this story?
- How does the story support the other parts of the tetrad (aesthetics, technology, gameplay)? Can it do a better job?
- How do the other parts of the tetrad support the story? Can they do a better job?
- How can my story be better?



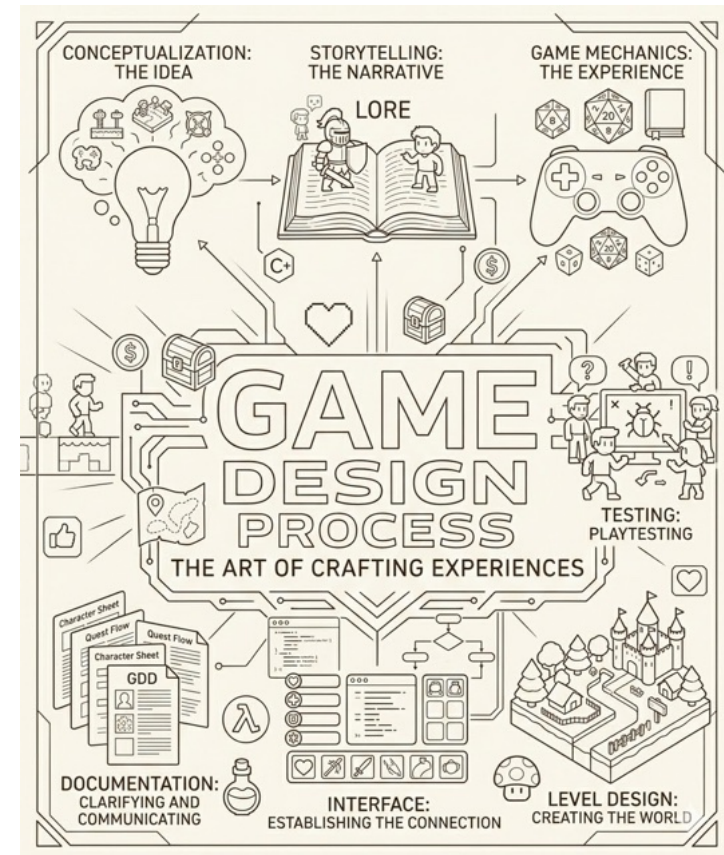
Storytelling

- Who is the game about?
- What is the goal?
- When does the story take place?
- Why is this going on in the story and why does the player care?
- How is the player going to accomplish the goals and interact with the story and the characters?



Game Design

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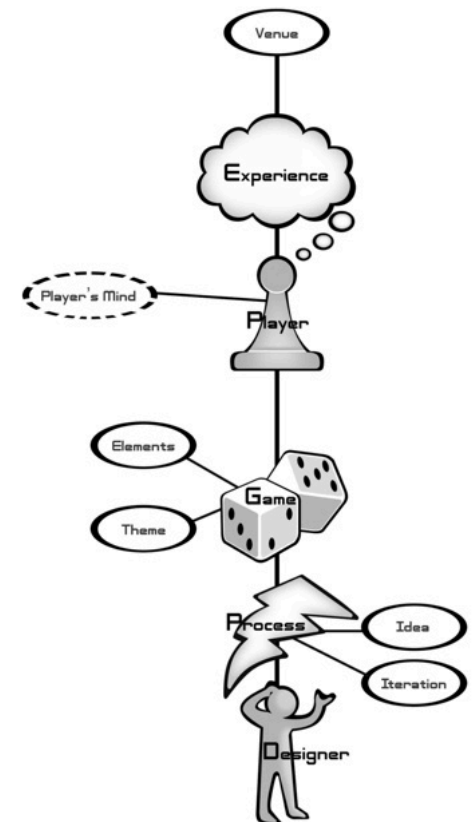
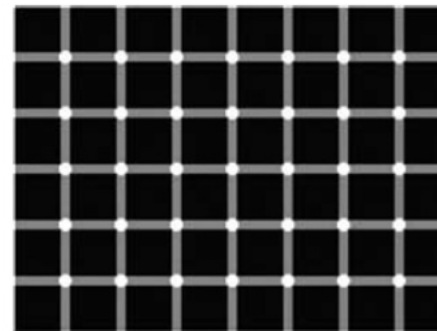
Game Mechanics

Game Mechanics

- Set of rules in a game that are intended to regulate and drive the gaming experience.
- Gameplay: Choices, challenges or consequences that players face while playing a game.
- Victory conditions: some games just get incredibly harder...
- Loss conditions: How a player loses.

Modeling the Experience

- The creation of the mechanics starts by modeling the experience of the player.
- A game does not need to emulate reality completely: A game is a simplified model of reality. Sometimes reality can't even be experienced.
- Games are created to capture the player's focus.



Focus

Proposed by psychologist Mihaly Csikszentmihalyi

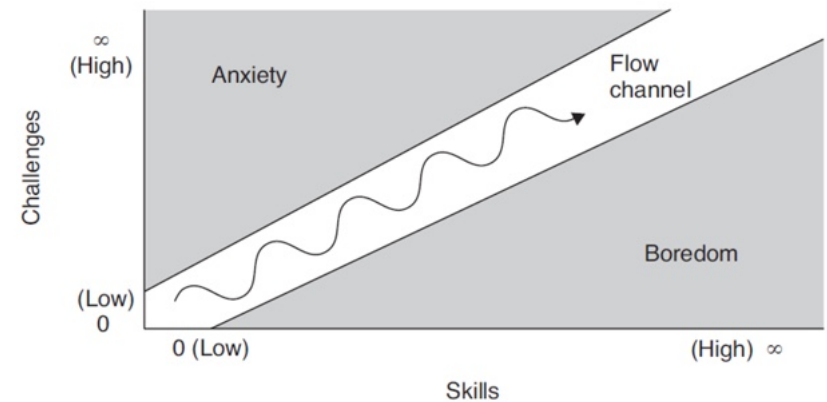
- Games should create a **State of Flow** using:

- Clear goals

State of sustained focus, pleasure and enjoyment.

- No Distractions
- Direct Feedback
- Continuous challenge

- Defined as “a feeling of complete and energized focus in an activity, with a high level of enjoyment and fulfillment.”



The Flow

- Does my game have clear goals? If not, how can I fix that?
- Are the goals of the player the same goals I intended?
- Are there parts of the game that distract players to the point they forget their goal? If so, can these distractions be reduced or tied into the game goals?
- Does my game provide a steady stream of not-too-easy, not-too-hard challenges, taking into account the fact that the player's skills may be gradually improving?
- Are the player's skills improving at the rate I had hoped? If not, how can I change that?



Game Mechanics (I)

- Game mechanics involves a series of challenges:
 - **Explicit:** intentional, immediate, and often intense.
 - * Player must react immediately with an action such as jump or turn.
 - **Implicit:** emergent feature of the game, not specifically added to the game.
 - * Determine how to divide and manage resources, deploy units in strategy games, etc.

“Game mechanics are the core of what a game truly is. They are the interactions and relationships that remain when all of the aesthetics, technology, and story are stripped away.”

Game Mechanics (II)

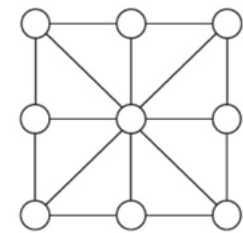
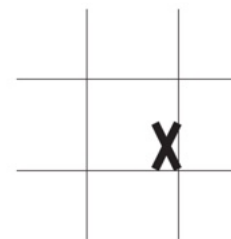
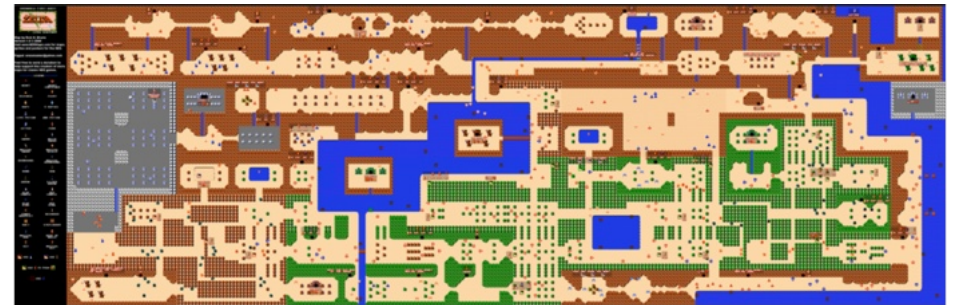
- Game mechanics and information:
 - **Perfect:** Logical challenge (e.g. chess)
 - **Imperfect:** Inference (appeal to curiosity) (e.g. card games, mastermind)
- Game mechanics and knowledge:
 - **Intrinsic:** knowledge is gained from within the game world (player's memory)
 - **Extrinsic:** relies on general knowledge from reality (ex. Who wants to be a millionaire? Trivial Pursuit...)

Jesse Schell's Taxonomy

- Game mechanics:
 - Mechanic 1: Space
 - Mechanic 2: Objects, Attributes and States.
 - Mechanic 3: Actions
 - Mechanic 4: Rules
 - Mechanic 5: Skill
 - Mechanic 6: Chance

Mechanic 1: Space

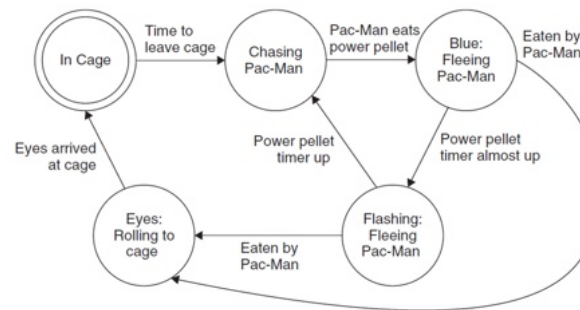
- The space is the “**magic circle**” of gameplay.
- It defines the various places that can exist in the game.
 - Discrete or Continuous
 - N Dimensional
 - M Bounded Areas (connected or not)



Mechanic 2: Objects, Attributes and States

- The game needs **Objects**, Characters, props, tokens, scoreboards, etc.
- **Objects** have **Attributes**: name, length, size, speed, force, id, tag, etc.
- **Attributes** have **States**: in a race game the speed changes state when we accelerate, brake or steer.

- Objects are *nouns*



[Pac Man state machine]

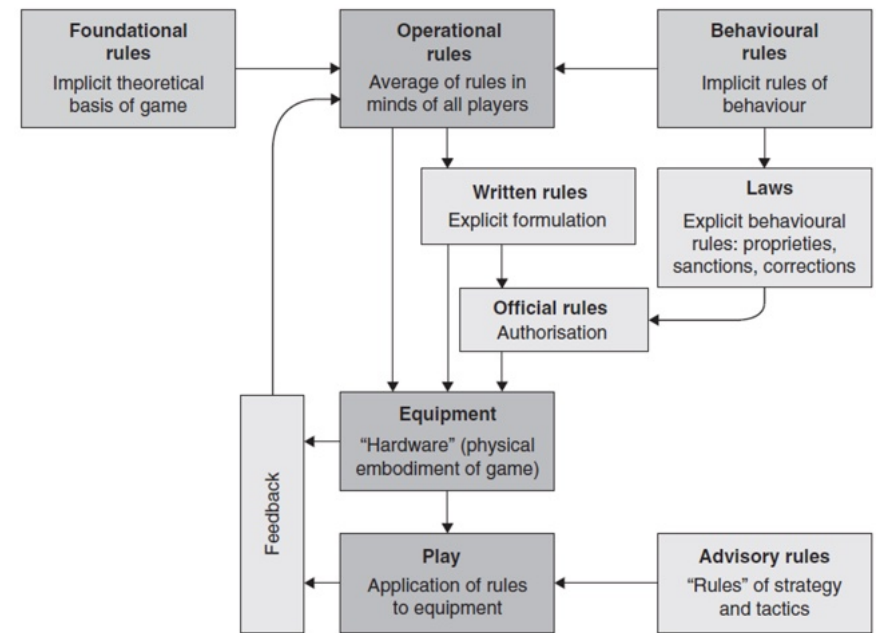


Mechanic 3: Actions

- Actions are *verbs* that can act on objects
- **Operative Actions:** basic actions the player can take
 - * “Move a checker forward”
- **Resultant Actions:** complex actions that combine several Operative Actions
 - * “Protect a checker from being captured by moving another checker behind it”

Mechanic 4: Rules

- The most fundamental mechanic
- Define space, objects, actions, consequences, constrains and goals

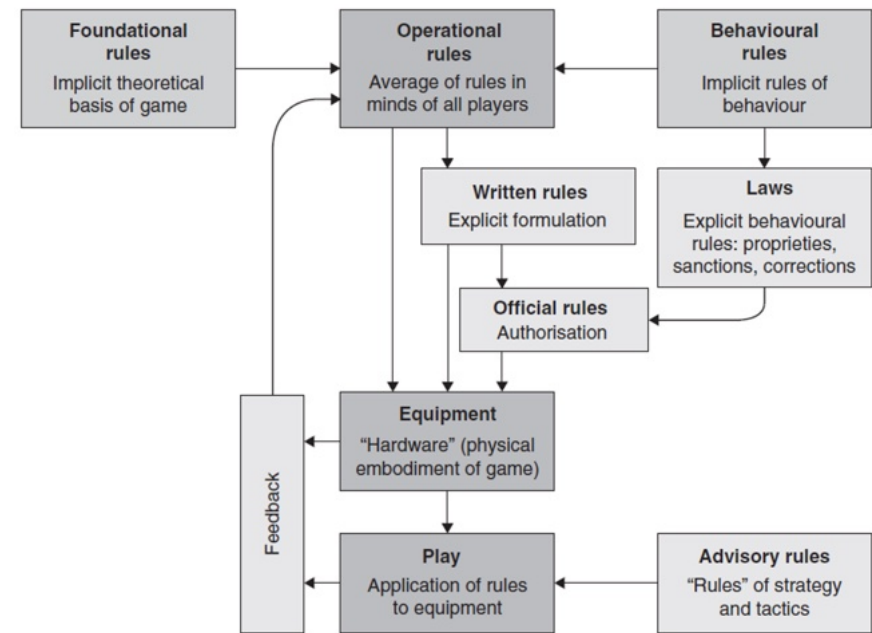


Parlett's Rule Analysis

Mechanic 4: Rules

- Parlett's Rule Analysis:

- Operational rules:** what players do to play the game
- Foundational rules:** are a mathematical representation of game state and how and when it changes
- Behavioral rules:** the implicit rules of the game.
- Written rules:** the documented rules of the game
- Laws:** also called tournament rules, they formalize what is fair to play.
- Official rules:** the laws of the tournament merge with the written rules to generate official rules.
- Advisory rules:** strategy to improve performance in the game.



Parlett's Rule Analysis

Mechanic 5: Skill

- The game may require certain skills from the player
 - Physical
 - Mental
 - Social



[Guitar Hero, 2005]

Mechanic 6: Chance

- Chance is an essential part of a fun game because chance means uncertainty, and uncertainty means surprises
- Depends on the study of Probabilities!



Game Balance (I)

- Game Mechanics should be balanced using several criteria.
- **Static Balance:** initial rules
- **Dynamic Balance:** balance can evolve over time

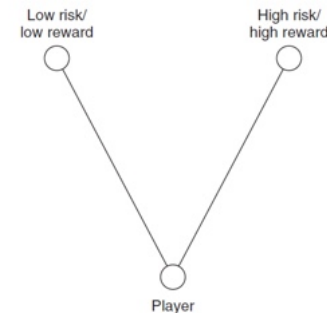
Game Balance (II)

1. Fairness

- Symmetrical Games: all players have the same resources
- Asymmetrical Games: might be desirable to simulate real world, personalization, to level the playing field (handicap)...

2. Challenge vs Success

3. Meaningful Choices: Choice should be balanced in number and reward



Game Balance (III)

4. Skill vs Chance
5. Head (thinking) vs Hand (interacting)
6. Competition vs Cooperation
7. Short vs Long Game
8. Rewards Economy (Points, Praise, Prolonged Play...)
9. Punishment (some is good, too much...)

Game Balance (IV)

10. Freedom vs Controlled Experience

11. Simple vs. Complex

12. Detail vs Imagination (Reality vs Player's mind)

Game Mechanics

- **Game Mechanics** include rules of play, gameplay elements and victory and loss conditions.
- The **state of flow** should be introduced while experiencing the game mechanics.
- Game Mechanics involve **challenges**, **information** access and possible prior **knowledge**.
- **Balance** should be provided either statically (starting conditions) or dynamically (as the game progresses).

